

Armaan Sandhu

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Education

Worcester Polytechnic Institute <i>Worcester, MA</i>	2025 - Present
Computer Science GPA: 4.0 Dean's List (Fall 2025, Spring 2026) Expected May 2029	
Carmel Junior College <i>Jamshedpur, India</i>	2010 - 2025
High School Diploma 97% (Unweighted) Valedictorian, Class of 2025	

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript

Tools: Git, GDB, Linux, Bash, Valgrind, llama.cpp, Redis, PostgreSQL, SQLite

Concepts: OS Kernel Development, Systems & Concurrent Programming, Binary Analysis, Query Planning, AI Infrastructure

Experience

Founding Engineer & Product Head <i>IMB Spectral</i>	June 2026 - Present
- Architecting a multi-model LLM routing platform: native provider integration layer, agentic tooling, model-testing playground, and credits-based billing. - Integrated TokenDrift as a core internal service for cross-provider tokenizer drift detection and mcp-persist as a core backbone for persistent context memory; designed HuggingFace Inference Provider integration and overall API surface.	
Open Source Author & Maintainer <i>mcp-persist</i>	May 2026 - Present
Built and maintain mcp-persist (10k PyPI downloads in the first month): production-grade MCP persistence layer across Redis, SQLite, and PostgreSQL with real-time streaming, PersistenceProxy, cross-backend migration, and a 300+ test suite; deployed in production by external teams.	

Open Source Contributions

lance / lancedb (Rust, Python)	May 2026 - June 2026
- Extended merge_insert to accept native DataFusion expressions directly, eliminating the SQL serialize-deserialize round-trip for programmatic predicate construction. - Fixed a critical data-loss bug in merge_insert where all-null values in the leading merge column silently dropped all matching rows in production. - Resolved thread-pool contention under concurrent search by moving blocking operations to a dedicated executor.	
LiteLLM Proxy (Python)	May 2026 - June 2026
- Fixed a slot leak in max_parallel_requests where cancelled streaming requests never released their slot, causing indefinite 429s under load. - Added Anthropic-native /v1/models content negotiation so Claude Code can use LiteLLM as a gateway; previously the model picker was empty despite /v1/messages working. - Fixed silent stripping of cache_control through OpenAI-compatible endpoints, preventing prompt caching from being silently disabled on every proxied request.	
Lightpanda Browser Engine (Zig)	May 2026 - June 2026
Implemented the W3C File API from specification and full file input support with CDP-based programmatic file injection, enabling automated upload workflows in a production headless browser.	
Other Contributions: systemd-sysupdate reboot/pending logic (<i>reviewed by Lennart Poettering</i>), CPython Cygwin build portability	

Projects

RunixOS (Rust)	Early 2026 - Present
Capability-based OS kernel for x86-64 with unforgeable-token IPC, ring-3 preemptive multitasking, and a novel TOCTOU finding: switching from cooperative to preemptive scheduling exposed a capability validation race that cooperative scheduling had been silently masking.	
TokenDrift / rag-timetravel (Python)	June 2026
TokenDrift diffs tokenizer behavior across LLM versions and providers; rag-timetravel is an event-sourced time-travel debugger for RAG pipelines built on LanceDB versioning.	

Awards & Recognition

- Valedictorian** | Carmel Junior College, Class of 2025
- Most Outstanding Student** | Carmel Junior College - excellence in academics, leadership, and extracurriculars
- National Gold Medalist (x2)** | Shotokan Karate India (IOC-recognized federation)
- International Silver Medalist & Bronze Medalist** | Shotokan Karate India
- Ni Dan (2nd Degree Black Belt)** | Shotokan Karate India - 10+ years competitive and instructing
- Interschool Java Coding Competition** | 2nd Place (2023, 2024)